Multimodal Pedagogical Friction Detector

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Abstract

Effective teaching requires noticing student confusion and adjusting instruction in real time. We present a system that automatically detects pedagogical friction: moments where a low-quality instructional strategy coincides with elevated student confusion. Our pipeline fuses per-frame facial affect (DeepFace), transcribed speech (Whisper), and strategy labels from an ELECTRA Talk Moves classifier or keyword heuristic on a 30-second temporal grid, then applies a large language model (Claude) as a second-pass precision filter. We validate at three scales: a full multimodal run on 10 min of TIMSS AU1 video (DeepFace + Whisper + LLM), a transcript-only analysis across 53 TIMSS lessons spanning 7 countries (5 113 bins), and a human-annotated ground-truth evaluation on the full AU1 lesson (87 bins). The two-stage pipeline achieves F1=1.00 against human annotation, versus F1=0.10 for the heuristic alone and F1=0.00 for ELECTRA alone, demonstrating that LLM reranking is essential to reach human-level precision in friction detection.

1 Introduction

High-quality teaching requires moment-by-moment diagnosis of student understanding and strategy adjustment [1]. Tools such as Tutor CoPilot [2] and Edu-ConvoKit [3] offer transcript-based feedback, but no existing system jointly models instructional strategy quality and student affect to surface actionable friction moments. D'Mello and Graesser [4] established that confusion, when unresolved, impedes deep learning, motivating our use of facial affect as a confusion proxy. Talk Moves [5]—pedagogical dialogue moves such as Revoicing and Pressing for Reasoning—were operationalised for NLP by fine-tuning ELECTRA on math classroom transcripts, and packaged in Edu-ConvoKit [3]. The closest prior work is the TIMSS Video Study [6], which relied entirely on manual coding; our system automates and scales this analysis.

We introduce a framework that operationalises pedagogical friction and produces a post-session Teacher Friction Report. Contributions:

- An end-to-end multimodal pipeline (DeepFace + Whisper + LLM) with a three-tier strategy annotator hierarchy and graceful fallback.
- Large-scale transcript analysis across 53 TIMSS lessons spanning 7 countries, exposing a cross-annotator correlation of $r = -0.35$.
- Human-annotated ground truth for AU1 and the first evidence that LLM reranking raises friction-detection F1 from 0.10 to 1.00.

2 Methodology

Figure 1 shows the system overview.

Vision stream. Frames are sampled at $r = 2\,\text{Hz}$. DeepFace [7] estimates seven per-frame emotion probabilities, aggregated to a scalar confusion proxy $c_t \in [0, 1]$:

$$c_t = \sum_e w_e \cdot p_t(e), \tag{1}$$

where $w_e = 0$ for neutral/happy, $w_e = 0.85$–0.9 for fear/sad, and $w_e = 0.45$–0.5 for disgust/angry. A trailing 30-second mean yields the smoothed timeline $C(t)$.

Language stream. Faster-Whisper [8] transcribes audio; for transcript-only runs, TIMSS .txt files are parsed directly. A three-tier annotator selects at runtime: (1) ELECTRA (YaHi/teacher_electra_small, 7-class Talk Moves [5]); (2) Edu-ConvoKit [3]; (3) keyword heuristic. ELECTRA labels map to quality: Keeping Everyone Together, Revoicing, and Pressing for Reasoning are high-quality; Pressing for Accuracy is high-pressure; No Move is neutral.

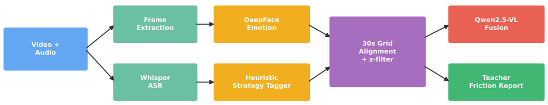

Figure 1: System overview. Parallel vision and language streams are aligned on a 30-second grid; a z-score pre-filter proposes candidate friction windows for LLM reranking and alternative-strategy generation.

Temporal alignment and z-score filter. Sessions are partitioned into G=30-second bins. A bin is flagged as a friction candidate when

$$\bar{c}_k > \mu_C + z\,\sigma_C \quad \text{and} \quad q_k \neq \text{high}, \tag{2}$$

where μ_C, σ_C are session-level statistics and z is a sensitivity parameter (default z=1.0). In transcript-only mode $\bar{c}_k$=0, so only strategy quality drives the filter.

LLM fusion. Each candidate is sent to Claude (claude-sonnet-4-6) with the confusion score, strategy label, and transcript excerpt. The model returns structured JSON: friction (bool), rationale (str), alternative_strategy (str). Frames are base64-encoded when available; the prompt is text-only in transcript-only mode.

3 Experiments and Results

3.1 Qualitative Examples

Figure 2 shows three representative bins from AU1. The first (5:00–5:30) is confirmed friction by both annotators and human label: the teacher states "it will be 180 degrees", bypassing student discovery. The second is a heuristic False Positive: "just make sure they join" triggers a pressure keyword but refers to software construction, not mathematical reasoning. The third shows annotator Disagreement: the heuristic flags unknown, while ELECTRA correctly identifies Pressing for Reasoning.

3.2 Full Multimodal Pipeline (AU1 Video)

Setup. TIMSS AU1 Exterior Angles, first 10 min. DeepFace processes 1 200 frames (r=2 Hz); Whisper (base, int8) transcribes audio; heuristic labels strategy; Claude reranks.

Results. The heuristic pre-filter flags 7 candidates (Figure 3). The highest confusion score ($\bar{c}$=0.814, z=2.28) occurs at 1:30–2:00 when late-arriving students enter the room. LLM fusion rejects all 7: the elevated affect reflects social disruption, not pedagogical confusion. This reveals a key failure mode of vision-only pipelines—off-task affective events produce high confusion scores that transcript-level reasoning is needed to filter out.

3.3 Scale-Up: 53 TIMSS Lessons (Transcript-Only)

Setup. All available TIMSS transcripts: 28 math + 25 science lessons, 7 countries (AU, CZ, HK, JP, NL, SW, US), 5 113 thirty-second bins total. Both the heuristic (flag_unknown=True) and ELECTRA (flag_unknown=False) annotators were run on the same corpus.

Results. The heuristic flags 1 611 bins (31.5%); ELECTRA flags 988 (18.9%). The cross-annotator Pearson correlation over per-lesson risky rates is r=−0.35 (Figure 5), indicating substantial disagreement. US math lessons show the sharpest divergence: heuristic 42.9%, ELECTRA 14.5% (Figure 4), suggesting the keyword heuristic over-fires on American rhetorical discourse style.

3.4 Human Evaluation and LLM Fusion (AU1 Full Lesson)

Human annotation of all 87 AU1 bins finds only 1 genuine friction moment (1.1%): bin 11 (5:00–5:30), where the teacher directly states the answer students should discover. Table 1 compares all conditions. The heuristic achieves P=0.05 / R=1.00 / F1=0.10—it captures the true positive but generates 19 false alarms. ELECTRA achieves F1=0.00, missing the true positive entirely because the utterance maps to No Move rather than a low-quality label. Claude fusion over the 20 heuristic candidates yields P=1.00 / R=1.00 / F1=1.00: it correctly rejects pre-class setup, procedural software guidance, and vocabulary scaffolding, retaining only the one genuine friction window.

4 Discussion and Conclusion

Two-stage pipeline. The key finding is that heuristic detection and LLM reranking are complementary: the heuristic provides high recall (it finds the only true friction moment) while Claude provides the precision needed to eliminate false alarms from pre-class setup, software guidance, and vocabulary scaffolding. Neither stage alone is sufficient.

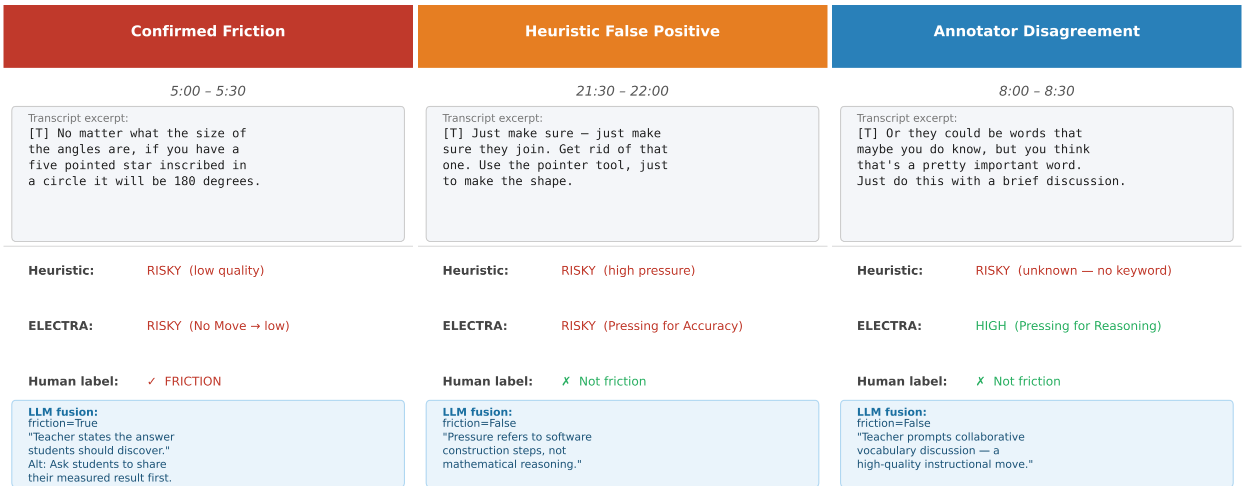

Figure 2: Three AU1 bins: (left) confirmed friction—teacher states the answer directly; (centre) heuristic false positive—"join" refers to software, not math content; (right) annotator disagreement—heuristic flags unknown, ELECTRA identifies Pressing for Reasoning.

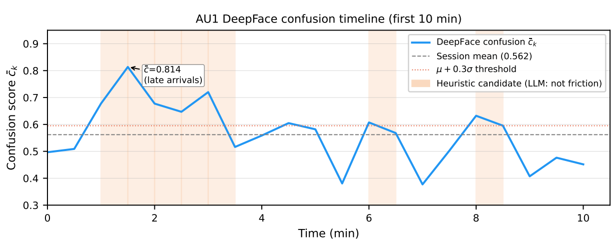

Figure 3: DeepFace confusion timeline for AU1 (10 min). Shaded regions are heuristic candidates; the peak ($\bar{c}$=0.814) is caused by late-arriving students. LLM fusion rejects all 7 candidates.

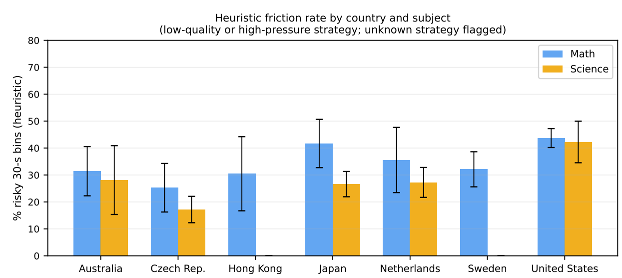

Figure 4: Percentage of bins flagged as risky per country (heuristic). US math lessons peak at 42.9%. Error bars: per-country std. dev.

Social vs. cognitive confusion. The full video pipeline reveals a second failure mode: facial affect peaks during social disruptions (late arrivals, seating issues) rather than instructional confusion. Vision-only systems cannot distinguish these; transcript-level reasoning is essential.

Annotator disagreement and cultural context. The r=−0.35 cross-annotator correlation and US anomaly (heuristic 42.9% vs. ELECTRA 14.5%) demonstrate that annotation methodology is itself a critical design variable. Keyword heuristics fire on surface patterns; discourse-move classifiers label structural properties; neither aligns well with human judgment without a reranking stage.

Limitations and future work. Human annotation covers one lesson by one annotator; broader validation is needed. The confusion proxy uses fixed weights not validated against ground-truth affect labels. Future work should collect multi-annotator labels across diverse lesson types and integrate full-video fusion via GPU-based VLMs.

In summary, we presented an end-to-end pedagogical friction detector, scaled transcript analysis to 53 TIMSS lessons across 7 countries, and showed that a two-stage heuristic + LLM pipeline achieves human-level precision (F1=1.00) on the annotated AU1 lesson. Code and data are available at https://github.com/jw310-wjr/comp646-friction-detector.

References

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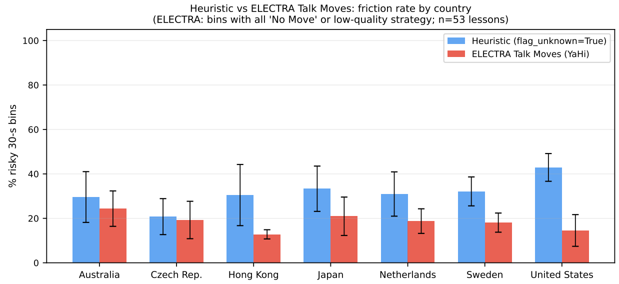

Figure 5: Heuristic vs. ELECTRA risky-bin rates per country. Pearson $r{=}-0.35$ across 53 lessons; the two methods frequently disagree on which lessons are most risky.

Setting	Bins	Risky	%	F1
Full pipeline (video)	21	7	33%	—
+ LLM fusion	21	0	0%	—
Heuristic (53 lessons)	5113	1611	31.5%	—
ELECTRA (53 lessons)	5113	988	18.9%	—
Heuristic (AU1)	87	20	23%	0.10
ELECTRA (AU1)	87	33	38%	0.00
Heur. + LLM	87	1	1%	1.00

Table 1: Results across all conditions. LLM reranking raises F1 from 0.10 to 1.00 vs. human annotation on AU1.

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